

The Dollar's Safe-Haven Status Across Two Shocks: Evidence from Liberation Day and the Strait of Hormuz Crisis

Bachelor's Thesis

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1 Introduction

Within a single year, two major shocks moved global markets in opposite directions for the U.S. dollar. On 2 April 2025—“Liberation Day”—the United States announced sweeping tariffs on virtually all of its trading partners, and the dollar *fell*: it depreciated against the euro, the yen and the Swiss franc even as equities sold off and volatility spiked, the textbook conditions under which the dollar normally *rises*. Less than a year later, as escalation in the Gulf threatened to close the Strait of Hormuz and sent crude sharply higher, the dollar did exactly what the safe-haven literature predicts—it appreciated. Two episodes, both with the United States implicated, produced a mirror image. This thesis asks why.

The puzzle speaks to a live debate about whether the dollar’s exceptional status is eroding. A growing literature argues that the currency’s safe-haven premium has weakened, pointing to the unusual co-movement of a falling dollar, rising Treasury yields and rising risk premia during the 2025 tariff episode ([Ostry, Lloyd and Corsetti 2025](#); [Kamin 2025](#); [Obstfeld and Zhou 2023](#); [Jiang et al. 2026](#)). Against this, the long tradition on the dollar’s “exorbitant privilege” and its role as the world’s reserve and safe-haven currency ([Gourinchas and Rey 2005](#); [Habib and Stracca 2012](#); [Ranaldo and Söderlind 2010](#)) implies the status should be durable. The two 2025–2026 shocks offer an unusually clean natural experiment for adjudicating the debate, because they differ in *kind*: one is a U.S. trade-policy shock, the other a geopolitical oil-supply shock.

Research question

The central question of this thesis is therefore: *does the dollar’s safe-haven response depend on the type of shock—even when the United States is, in some sense, the originator of both?* The design contrasts three episodes. The Liberation Day tariff announcement is a purely U.S.-originated economic-policy shock. The 2026 Israel–Iran/Hormuz crisis is a geopolitical oil-supply shock in which the United States is implicated as a co-belligerent but still functions as a financial safe harbour. And the 2022 Russian invasion of Ukraine—an external military/oil shock not authored by the United States—serves as a control for the “classical” safe-haven response. Comparing the three isolates whether the dollar’s behaviour is conditioned by the nature of the shock or merely by who caused it.

Approach and contribution

The thesis answers the question with two complementary empirical layers on a single daily dataset spanning January 2019 to June 2026. An *event study* measures the cumulative abnormal response of the dollar, a cross-section of currency baskets, gold, equities and crude oil around each shock, and tests formally whether the dollar’s response differs across events. A complementary *regression* then tests whether the dollar’s relationship with risk itself changed across the episodes—whether its safe-haven sensitivity to volatility held or broke. The two layers reinforce one another: the event study shows that discrete, dated shocks move the dollar in opposite directions depending on shock type, and the regression shows that the dollar’s safe-haven reflex broke down for the trade-policy shock yet held for the geopolitical ones.

The contribution is threefold. First, to the best of the author’s knowledge this is the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets. Second, the thesis bridges the safe-haven and commodity-currency literatures by carrying the oil channel and a gold benchmark through the same design, showing that gold behaves as a hedge *against* the dollar rather than as an unconditional substitute for it. Third, the rebuilt carry/dollar risk factors and the WMR currency panel extend standard cross-sectional tools through to mid-2026, allowing the most recent shocks to be studied with consistent, citable data.

The remainder of the thesis is organised as follows. Chapter 2 reviews the safe-haven, dollar-erosion and commodity-currency literatures and states the predictions that follow from them. Chapter 3 narrates the 2025–2026 events. Chapter 4 describes the data and the event-study and regression methodologies. Chapter 5 presents the event-study results and Chapter 6 the regression test. Chapter 7 discusses the findings and their implications, and Chapter 8 concludes.

2 Literature Review

This chapter situates the thesis within five strands of research—currency factor models, safe-haven currencies and gold, geopolitical risk, tariff shocks, and commodity currencies—and closes by distilling them into testable hypotheses. The unifying question is when, and through which channel, the U.S. dollar functions as a safe haven, and whether that function survives when the United States is itself the source of the shock.

2.1 Currency Returns and Factor Models

The modern cross-section of currency returns is organised around a small number of risk factors. [Lustig, Roussanov and Verdelhan 2011](#) identify a “slope” factor (the return on a portfolio long high-interest-rate and short low-interest-rate currencies) that explains most of the cross-sectional variation in carry returns and is tied to innovations in global equity-market volatility; high-interest-rate currencies load more heavily on this common factor, so investors in the carry trade are compensated for bearing global risk. This logic motivates sorting currencies into portfolios by their exposure to common shocks, and the dollar and carry factors derived from it underpin the portfolio construction used in Chapter 4. Related work on dominant-currency pricing links the pricing of trade and the currency composition of invoicing to these risk premia ([Dalgic and Ozhan 2024](#); [Dalgic and Ozhan 2025](#)).

The dollar occupies a special position in this system. [Gourinchas and Rey 2005](#) document the United States’ “exorbitant privilege”—a persistent excess return on its external assets over its liabilities—and [Gourinchas, Rey and Govillot 2017](#) show that this privilege is mirrored by an “exorbitant duty”: the hegemon transfers wealth to the rest of the world precisely during global crises, with transfers of roughly 19% of U.S. GDP during the 2007–2009 financial crisis. [Gourinchas, Rey and Sauzet 2019](#) embed this in a broader account of the international monetary system and a renewed Triffin dilemma, while [Obstfeld and Zhou 2023](#) describe a “global dollar cycle” in which dollar appreciation tightens financial conditions worldwide. Together these contributions frame the dollar’s safe-haven status as a structural feature of the international monetary system—and therefore something whose erosion would be economically consequential.

2.2 Safe-Haven Currencies and Gold

Following [Baur and Lucey 2010](#), an asset is a *hedge* if it is uncorrelated with risky assets on average and a *safe haven* if it is uncorrelated with (or negatively correlated with) them specifically during market stress; they find gold to be a hedge against equities on average and a safe haven in extreme declines, though the safe-haven property is short-lived. This conditional, crisis- contingent definition is the one adopted throughout the thesis. Applied to currencies, [Ranaldo and Söderlind 2010](#) show that the Swiss franc and Japanese yen appreciate against the dollar when U.S. equities fall and volatility rises, with the effect strengthening non-linearly in the volatility factor and during crises. [Bock and Carvalho Filho 2015](#) document the same recurrent pattern across risk-off episodes—the yen, franc and dollar all appreciate against other currencies—and relate it to fundamentals such as the nominal interest rate and the net foreign asset position. [Habib and Stracca 2012](#) push beyond the carry trade to ask what *makes* a currency a safe haven, finding that the net foreign asset position, low external vulnerability, and economic size are robustly associated with safe-haven status, whereas the interest-rate spread against the United States is not. Methodologically, the time-varying correlation approach used to test these properties—and applied to gold and other assets by [Kumar, Mohan and Niveditha 2025](#)—rests on the dynamic conditional correlation model discussed in Section 2.3 and Chapter 4.

2.3 Geopolitical Risk and Currency Markets

[Caldara and Iacoviello 2022](#) construct the news-based Geopolitical Risk (GPR) index used in this thesis, showing that it spikes around wars and major adverse events and that higher geopolitical risk foreshadows lower investment and employment and larger downside risks; importantly, they distinguish the *threat* of adverse events from their *realisation*, a distinction directly relevant to staged crises such as the Hormuz episode. Building on such measures, [Liu and Zhang 2024](#) study geopolitical-risk-sorted currency portfolios and report a positive risk premium for exposure to geopolitical shocks, and the [International Monetary Fund 2025](#) finds that geopolitical-risk shocks predict currency depreciation and equity declines, particularly in more vulnerable economies. This literature motivates treating geopolitical events as priced risk rather than noise, and supplies the GPR series used as an external control in the empirical chapters.

2.4 Tariff Shocks and Exchange Rates

The 2025 tariff episode has generated a fast-growing literature that is central to this thesis. [Kamin 2025](#) provides econometric evidence that, following the Liberation Day announcements of April 2025, the dollar switched from behaving as a safe haven—appreciating in periods of market volatility—to behaving as a “risk-on” currency that moves inversely with volatility. [Jiang et al. 2026](#) document the same break in historical correlations (the dollar depreciating while U.S. rates rose relative to the euro, the VIX increased, and the dollar convenience yield fell) and interpret it through shifts in global demand for dollar safe assets, calibrating a steady-state real dollar depreciation of around 7.6% should the United States lose reserve-currency demand. [Hassan et al. 2025](#) provide the theoretical counterpart: in a general-equilibrium model where currency safety is endogenous, a trade war that isolates U.S. goods markets erodes the dollar’s safety premium and raises U.S. interest rates. On the empirical exchange-rate response, [Ostry, Lloyd and Corsetti 2025](#) construct a tariff-shock database and show that exchange rates react to U.S. tariff shocks systematically once the size and relevance of the measures (and expected retaliation) are accounted for, while [Kalemlı-Özcan, Soylu and Yildirim 2026](#) emphasise the role of tariff *uncertainty* in the dollar’s behaviour. The theoretical tariff–exchange-rate offset is formalised by [Itskhoki and Mukhin 2025](#). Policy institutions corroborate the market dislocation: the [European Central Bank 2025](#) and the [European Central Bank 2026](#) document trade fragmentation, the pass-through of higher U.S. tariffs, and the effects of trade-policy uncertainty on activity. This thesis contributes a direct event-study comparison of this trade-policy shock against contemporaneous military/oil-supply shocks.

2.5 Commodity Currencies and Oil

[Chen and Rogoff 2003](#) establish that the U.S.-dollar price of commodity exports exerts a strong and stable influence on the real exchange rates of commodity exporters such as Australia, Canada and New Zealand, providing the basis for the oil-exporter versus oil-importer portfolio split used in Chapter 4. The nature of the oil shock matters, however: [Kilian 2009](#) shows that oil-price movements driven by supply disruptions, aggregate demand, and precautionary (oil-specific) demand have distinct macroeconomic consequences, so a price increase caused by a supply scare is not equivalent to one caused by strong demand—nor to a tariff-induced demand collapse.

This typology maps directly onto the thesis’s contrast between the demand-side tariff shock (falling oil) and the supply-side Hormuz shock (surging oil). For the 2026 episode specifically, [Fattouh and Economou 2026](#) provide a detailed anatomy of the Strait of Hormuz oil shock and the associated disruption to Gulf output, while [Kilian et al. 2026](#) quantify the inflationary consequences of the 2026 Iran war through oil prices, and the [International Monetary Fund 2026](#) documents the regional and global spillovers, including Brent rising above \$100 per barrel and the role of the ceasefire in de-escalation.

2.6 Hypotheses

Taken together, these strands point to a single organising prediction: the dollar’s safe-haven response is conditional on the *type* of shock rather than on whether the United States is its originator. Both the external benchmark (the 2022 Russian invasion of Ukraine) and the U.S.-originated Hormuz crisis are military/oil-supply shocks, around which the safe-haven literature (Section 2.2) implies the dollar should appreciate; the Liberation Day tariff announcement, by contrast, is a U.S. trade-policy shock that the dollar-erosion literature (Section 2.4) suggests should weaken the very currency it originates from. Gold offers a natural counterpoint: as an asset external to any sovereign it might be expected to bid in *both* kinds of episode, so that comparing it against the dollar isolates what is specific to the dollar’s status ([Baur and Lucey 2010](#))—a prediction this thesis puts directly to the data. The cross-section of currencies should in turn sort on the oil channel, exporter currencies outperforming importers when the Hormuz supply shock lifts crude but losing ground when the tariff shock depresses oil through weaker demand ([Chen and Rogoff 2003](#); [Kilian 2009](#)).

The remainder of the thesis puts these predictions to the data: the dollar and the oil cross-section in the event study of Chapter 5, with gold examined alongside them, while a regression test in Chapter 6 establishes whether the dollar’s safe-haven relationship with risk held or broke in each episode.

3 Background: The Events of 2022–2026

This chapter narrates the episodes the empirical chapters analyse and fixes the dates used throughout. Three are central—the 2025 U.S. tariff shock, the 2026 Iran/Hormuz crisis, and the 2022 Russian invasion of Ukraine as an external control—together with two within-crisis sub-events (the April 2025 tariff pause and the June 2026 ceasefire) that serve as natural reversals.

3.1 Liberation Day and the U.S. Tariff Shock

On 2 April 2025, branded “Liberation Day,” the United States announced broad tariffs on nearly all trading partners. The market reaction was severe and, for the dollar, anomalous. Equities fell sharply, implied volatility spiked, and crude oil dropped as markets priced weaker global demand—yet the dollar *depreciated* rather than attracting the usual flight-to-safety bid, sliding several percent against the euro, yen and Swiss franc. Long-term Treasury yields, after an initial decline, backed up even as the currency weakened, a co-movement widely read as evidence that the episode dented the dollar’s safe-haven premium rather than reinforcing it ([Ostry, Lloyd and Corsetti 2025](#); [European Central Bank 2025](#)). A partial ninety-day tariff pause on 9 April 2025 tempered the equity drawdown but did not restore the dollar, suggesting markets treated the shock as a regime change in U.S. trade policy rather than a transient surprise.

3.2 The Israel–Iran Escalation and the Strait of Hormuz

The Gulf episode unfolded in stages. A first, twelve-day confrontation in June 2025 (Israeli and U.S. strikes on Iranian facilities, an Iranian response, and a rapid ceasefire) lifted crude only briefly and left currencies little changed. Escalation resumed in February 2026 and turned acute on 4 March 2026, when Iran moved to close the Strait of Hormuz—the conduit for roughly a fifth of seaborne oil—triggering attacks on shipping and a sharp spike in Brent ([Fattouh and Economou 2026](#); [Kilian et al. 2026](#)). The dollar strengthened on impact, the classical safe-haven response and the mirror image of the tariff episode. A further round of U.S. strikes followed on 28 May 2026, and a sixty-day ceasefire announced on 11 June 2026, committing to reopen

the Strait, reversed much of the move in both crude and the dollar—providing a clean within-crisis test of whether the premium that the threat put into the dollar came back out on de-escalation.

3.3 The Ukraine Control

The Russian invasion of Ukraine on 24 February 2022 is the external benchmark: a military/oil-supply shock not authored by the United States. It produced the textbook pattern—a sustained dollar appreciation alongside a jump in energy prices—and so anchors what a “classical” safe-haven response looks like when the United States is a bystander rather than a protagonist.

3.4 Event Timeline

Table 3.1: Event timeline and shock classification

Date	Event	Type	Oil
2022-02-24	Russia invades Ukraine	External military/oil (control)	↑
2025-04-02	Liberation Day tariffs	U.S. trade policy	↓
2025-04-09	90-day tariff pause	sub-event	↓
2025-06-13	Twelve-day war (Israel/U.S.–Iran)	U.S.-implicated military	↑ (mild)
2026-02-28	Hormuz crisis begins	U.S.-implicated oil supply	↑↑
2026-03-04	Strait of Hormuz closed	sub-event	↑↑
2026-05-28	U.S. strikes on Iran	U.S.-implicated military	↑
2026-06-11	Sixty-day ceasefire	de-escalation	↓

Notes. “Type” classifies each shock by origin and nature; the final column gives the qualitative crude-oil response. Sub-events are reversals nested within their parent episode.

3.5 Key Variables Over the Sample

Figure 3.1 plots the broad dollar index, the VIX, crude oil and the geopolitical-risk index over 2019–2026, with the four headline events shaded. The visual record previews the central result: the dollar rises into the Ukraine and Hormuz bands but falls through the Liberation Day band, while oil and risk gauges move with the geopolitical episodes. The formal analysis follows in Chapters 5 and 6.

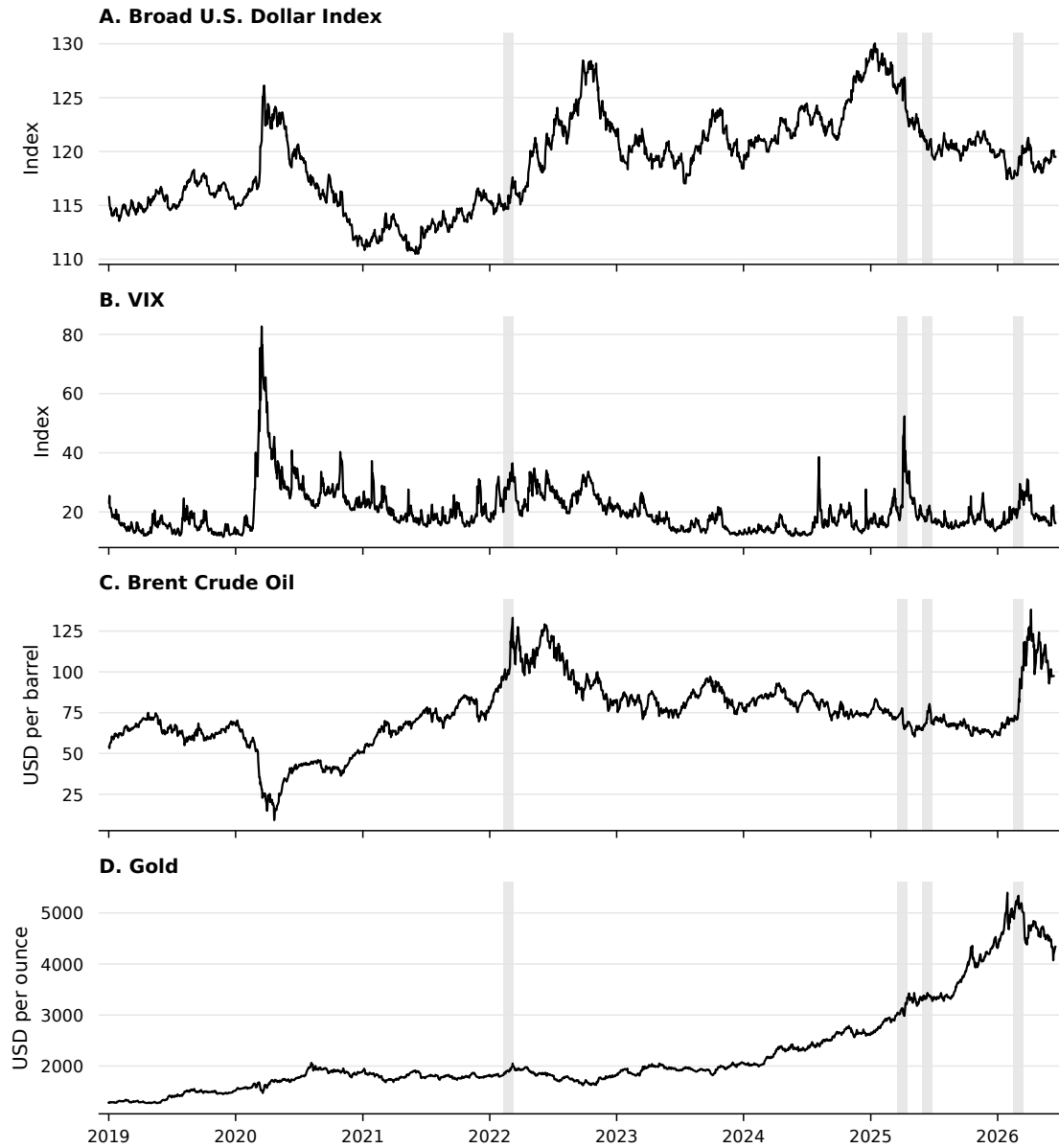


Figure 3.1: Key variables over the sample with the four main events shaded (Ukraine invasion, Liberation Day, twelve-day war, Hormuz crisis). Sources: FRED, LSEG, and Caldara and Iacoviello's Geopolitical Risk index ([Caldara and Iacoviello 2022](#)).

4 Data and Methodology

4.1 Data

4.1.1 Exchange Rates and Financial Variables

The currency panel comprises 32 exchange rates against the U.S. dollar, sampled daily from 1 January 2019 to 30 June 2026. Spot rates are taken from LSEG/Refinitiv at the WM/Reuters (WMR) fixing and exported as a single consistent panel; the Federal Reserve’s H.10 release (via FRED) serves as a cross-check. All rates are expressed as foreign currency per U.S. dollar, so that a rise denotes dollar appreciation; the four rates LSEG quotes in the opposite convention (EUR, GBP, AUD, NZD) are inverted accordingly. Daily returns are log changes, $r_{i,t} = \ln(S_{i,t-1}/S_{i,t})$, signed so that a positive value is an appreciation of the foreign currency against the dollar.

The macro-financial block is drawn from FRED: the broad, advanced-economy and emerging-market nominal dollar indices (DTWEXBGS, DTWEXAFEGS, DTWEXEMEGS); the 5- and 10-year Treasury yields (DGS5, DGS10), the 5-year TIPS yield (DFII5) and 5-year breakeven inflation (T5YIE); the VIX (VIXCLS); Brent and WTI crude (DCOILBRENTU, DCOILWTIC0); Henry Hub natural gas (DHHNGSP); and the S&P 500 (SP500). Three series not carried by FRED are taken from LSEG: the gold spot price (XAU=ZKBZ, USD/oz), the Euro Stoxx 50 price index (.STOXX50E, EUR) and the Dutch TTF front-month natural-gas contract (TRNLTTFMc1, EUR/MWh). Indices, equities and commodities enter as log returns; yields and the VIX enter in first differences.

4.1.2 Geopolitical and Policy Uncertainty Indices

Geopolitical risk is measured by the Caldara and Iacoviello (2022) Geopolitical Risk (GPR) index, used at daily frequency—with its *threat* and *act* sub-indices—and monthly, downloaded from the authors’ site. Policy uncertainty is captured by the U.S. daily Economic Policy Uncertainty (EPU) index and the Trade Policy Uncertainty (TPU) index from *policyuncertainty.com*. The dollar and carry risk factors are reconstructed from spot and one-month forward exchange rates following Lustig, Roussanov and Verdelhan (2011): the one-month forward discount $f_{i,t} - s_{i,t}$ proxies the interest-rate differential under covered interest parity; currencies are sorted on this signal each month, and the dollar factor (the average forward-implied excess return) and the carry factor (the high-minus-low portfolio return) are formed.

Rebuilding the factors from current market data extends them through the 2026 sample, replacing the course-supplied series that end in 2017.

4.1.3 Portfolio Construction

Two basket schemes are used. The first sorts currencies by their average one-month forward discount over the sample into terciles: the low-yield tercile forms the *safe* (funding) basket and the high-yield tercile the *risky* (carry) basket, with pegged and managed currencies (DKK, HKD, SAR, KWD, CNY) excluded from the sort. This is the classification that underlies the carry factor and the main event-study specification (Figure 4.1). Because the low-yield tercile also contains low-rate Asian currencies that are not conventional safe havens, a second, literature-standard scheme is reported as a robustness check: a *named-haven* safe basket of the yen, Swiss franc and euro against a *risky* basket of the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah. Independently, currencies are split by net oil-trade position into oil exporters (NOK, CAD, MXN, COP, BRL, SAR, KWD) and oil importers (JPY, KRW, INR, THB, TWD, TRY). All baskets are equally weighted; the equal-weighted average of all currency returns defines the dollar basket (USD_EW).

4.2 Event Study Methodology

The main specification is the *constant-mean return* model. For each series i the normal return is its average over an estimation window of $[-130, -11]$ trading days preceding the event; the abnormal return on day t of the event window is the deviation from that mean,

$$AR_{i,t} = R_{i,t} - \hat{\mu}_i, \quad \hat{\mu}_i = \frac{1}{120} \sum_{s=-130}^{-11} R_{i,s}, \quad (4.1)$$

and the cumulative abnormal return aggregates these over the window,

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{i,t}. \quad (4.2)$$

The event window is $[-20, +20]$ trading days. Cumulative windows $CAR(0, h)$ for $h \in \{1, 5, 10, 20\}$ capture the announcement reaction, while a long horizon $CAR(0, 50)$ —estimated on $[-170, -51]$ so the estimation window does not overlap the wider

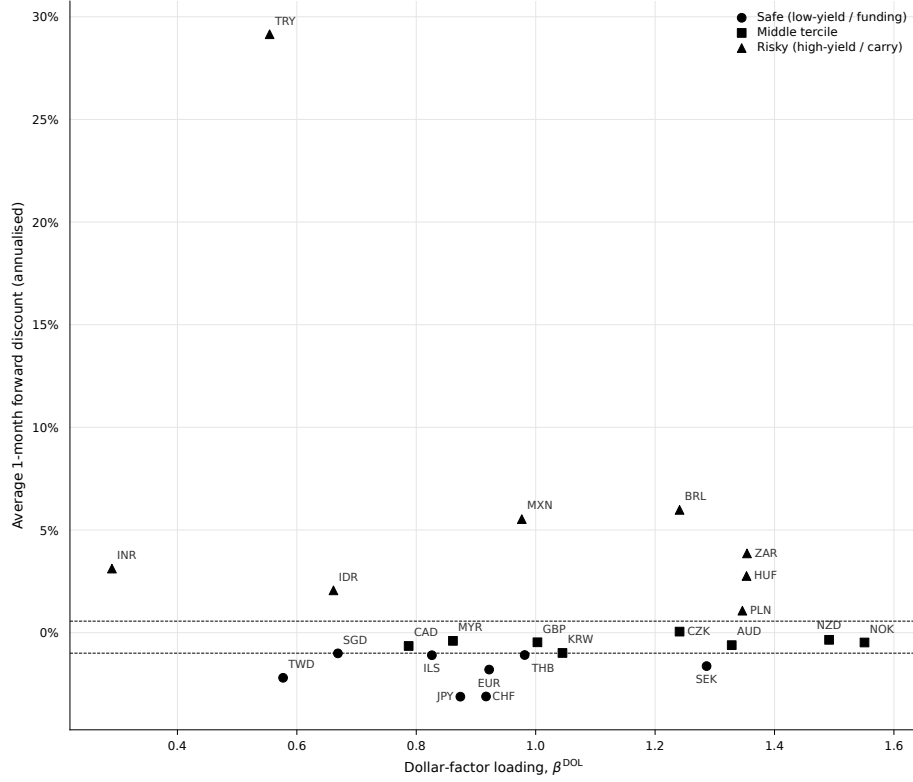


Figure 4.1: Currency classification. Each currency is plotted by its dollar-factor loading β^{DOL} (from a time-series regression of its monthly excess return on the reconstructed dollar and carry factors) against its average one-month forward discount, the signal used for the tercile sort. Dashed lines mark the tercile cut-offs, so marker shape (bucket) and vertical position agree: low forward discount $\rightarrow$ safe (funding), high $\rightarrow$ risky (carry).

event window—gauges persistence. The significance of a single CAR is assessed with $t = \text{CAR}/(\hat{\sigma}_i\sqrt{L})$, where $\hat{\sigma}_i$ is the estimation-window standard deviation of AR_i and $L = t_2 - t_1 + 1$. The difference between two events' CARs is tested with $\Delta = \text{CAR}_A - \text{CAR}_B$ and standard error $\sqrt{L(\hat{\sigma}_A^2 + \hat{\sigma}_B^2)}$, treating the (non-overlapping) event windows as independent.

The constant-mean model is preferred to the market model because the object of study is the dollar itself. The natural market factor for the currency baskets is the dollar factor, which is endogenous to the response under examination: regressing dollar exchange rates on a dollar factor would absorb the common dollar movement into $\beta_i R_{m,t}$ and leave little in the abnormal return. As a robustness check on *individual* currencies, the market model

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}, \quad \text{AR}_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}), \quad (4.3)$$

with $R_{m,t}$ the equal-weighted dollar factor, is reported in Appendix 10 and leaves the safe-haven hierarchy unchanged. Where cross-sectional dependence across currencies is a concern, the Kolari and Pynnönen (2010) adjusted test can replace the plain t -test; given the small number of non-overlapping events, that refinement is relegated to the appendix.

4.3 A Regression Test of the Safe-Haven Breakdown

The event study measures *whether* the dollar moved; a complementary regression measures *whether its relationship with risk changed*. The classical safe-haven property is that the dollar appreciates when global risk rises, so the daily broad-dollar return is regressed on the daily change in the VIX, interacted with a dummy for each crisis window:

$$r_t^{\text{USD}} = a + b_1 \Delta \text{VIX}_t + \sum_k \beta_k (\Delta \text{VIX}_t \times D_t^k) + \sum_k \gamma_k D_t^k + \varepsilon_t, \quad k \in \{\text{tariff, Hormuz, Ukraine}\}, \quad (4.4)$$

where D_t^k equals one from the event day through twenty trading days after shock k . A positive ΔVIX is a risk-off move and a positive r^{USD} a dollar appreciation, so $b_1 > 0$ is the normal safe-haven reflex; a negative interaction β_k means the reflex weakened during episode k , and the dollar's risk-sensitivity within that window is $b_1 + \beta_k$. Standard errors are Newey–West (HAC, five lags), robust to the volatility clustering of daily returns. Results are reported in Chapter 6.

5 Event Study Results

This chapter measures how the U.S. dollar, the cross-section of currency baskets, and crude oil responded to four shocks. The design isolates the thesis’s central contrast: three of the shocks are *U.S.-originated* (the Liberation Day tariff announcement, the twelve-day war, and the 2026 Hormuz crisis), while the 2022 Russian invasion of Ukraine is an *external* military/oil shock in which the United States is not the instigator, and which therefore serves as a benchmark for the “classical” safe-haven response. Abnormal returns are computed with a constant-mean model: the normal return of each series is its mean over the estimation window $[-130, -11]$ trading days, the abnormal return is the deviation from that mean, and the cumulative abnormal return $CAR(0, h)$ sums abnormal returns from the event day through trading day h .¹ Significance is assessed with a t -test on the CAR, $t = CAR/(\hat{\sigma}_{est}\sqrt{L})$. Tables 5.1 and 5.2 collect the headline numbers; the discussion below reads them event by event.

Table 5.1: The U.S. dollar’s response by event and horizon

Event	Broad USD index — $CAR(0, h)$, %				
	$h=1$	$h=5$	$h=10$	$h=20$	$h=50$
<i>Panel A. U.S. trade-policy shock</i>					
Liberation Day	−1.41***	−0.00	−2.81**	−3.62**	−6.34***
Tariff pause	−1.46***	−3.01***	−3.32***	−4.29***	−6.74***
<i>Panel B. U.S. military / oil-supply shock</i>					
Twelve-day war	+0.06	+0.94	−0.03	+0.55	−0.12
Hormuz crisis	+1.40***	+1.54***	+2.12***	+3.29***	−0.13
Hormuz ceasefire	−1.18***	−1.63***	−1.41*	−1.81	+0.71
<i>Panel C. External control (non-U.S.)</i>					
Ukraine invasion	+0.50	+1.06*	+1.70**	+0.94	+3.12**

Notes. Cumulative abnormal returns of the broad nominal U.S. dollar index (FRED DTWEXBGS), in percent, from a constant-mean model. Estimation window $[-130, -11]$ trading days for $h \leq 20$; $[-170, -51]$ for $h=50$ (pushed back so it does not overlap the wider event window). Positive values denote dollar appreciation. Tariff pause and Hormuz ceasefire are sub-events within their parent windows. The US strikes (May 2026) and the June 2026 ceasefire are omitted because the sample ends 2026-06-16, leaving too few post-event days for a stable window. *, **, *** denote significance at the 10%, 5% and 1% level.

¹The constant-mean specification is preferred to the market model here because the object of study is the dollar itself: the natural “market” factor for the currency baskets is the dollar factor, which is endogenous to the response we seek to measure and would absorb the common dollar movement into $\beta R_{m,t}$. The market model with the dollar factor is therefore reported only as a robustness check on *individual* currencies (Appendix 10). See Section 4.2.

5 Event Study Results

Table 5.2: Cross-section of five-day responses: the mirror image

Series, CAR(0, 5) (%)	U.S.-originated			Control
	Liberation Day	Hormuz	Twelve-day war	Ukraine
Broad USD index	−0.00	+1.54***	+0.94	+1.06*
Dollar (equal-weighted)	+0.26	+1.39**	+0.89	+1.04
Safe currencies (carry)	+0.97	−1.03	−1.08	−1.04*
Risky currencies (carry)	−0.84	−1.88***	−0.91	−2.34*
Oil exporters	−0.90	−0.71	−0.49	+0.25
Oil importers	+0.85	−1.10**	−1.19	−0.72
Gold	−1.56	−4.22	−1.60	+1.34
Euro Stoxx 50	−14.58***	−8.23***	−2.79	−6.03**
Brent crude	−14.24***	+27.83***	+11.21***	+13.63***
WTI crude	−13.19***	+34.71***	+10.34**	+14.26***
<i>Robustness: named safe-haven baskets</i>				
Safe (JPY, CHF, EUR)	+2.02*	−1.25	−1.45	−0.68
Risky (TRY, BRL, ZAR, MXN, COP, IDR)	−1.77*	−1.24	−0.37	+0.20

Notes. Five-day cumulative abnormal returns (%), constant-mean model, estimation window $[-130, -11]$. Liberation Day is the trade-policy shock; Hormuz and the twelve-day war are the military/oil-supply shocks; Ukraine is the non-U.S. control. For the currency baskets a positive value means the basket *appreciated against the dollar*; for the dollar indices, gold and equities positive means a price increase, and for crude positive means a price increase. *Safe/Risky (carry)* are the bottom and top terciles of the rebuilt forward-discount classification (Section 4.1.3); the robustness panel reports the literature-standard named-haven baskets. Gold's CARs are baseline-sensitive given its exceptional 2025–26 trend and are read for sign rather than magnitude (Section 5.4). *, **, *** denote significance at the 10%, 5% and 1% level.

5.1 CARs: Tariff Shock (Liberation Day)

The April 2, 2025 tariff announcement triggered an immediate and *persistent* depreciation of the dollar. The broad dollar index fell 1.41% on the announcement day (significant at the 1% level) and continued to slide: -2.81% by ten trading days, -3.62% by twenty, and -6.34% by fifty (Table 5.1, Panel A). The 90-day tariff pause on April 9 did not arrest the move—if anything it deepened it, the broad index reaching -6.74% fifty days out—indicating that markets read the episode as a durable shift in the U.S. policy regime rather than a transient shock.

The cross-section confirms that the safe-haven premium did not accrue to the dollar but *leaked away from it*. Over the five-day window the named-haven basket—the yen, Swiss franc and euro—appreciated 2.02% against the dollar, the move concentrated in the franc ($+3.08\%$, significant at the 1% level); the broader carry-sorted safe basket, which also contains low-yield Asian currencies, gained a more muted 0.97% (Table 5.2). Either way the flight to quality went to competing havens rather than to the dollar. Crude oil fell sharply—Brent -14.24% and WTI -13.19% over five days—European equities sold off hard (Euro Stoxx 50 -14.58%), and the oil-exporter basket slipped 0.90%, consistent with the tariff shock operating through a global *demand*-destruction channel rather than a supply channel. This is the signature of the United States acting as a destabiliser: a U.S.-made shock that simultaneously weakens the dollar, depresses oil and equities, and redirects safe-haven flows to competing currencies.

5.2 CARs: Iran/Hormuz Shock

The two military/oil-supply episodes are reported separately because they differ in both intensity and the role the data assign to the United States.

Twelve-day war (June 2025). The response is muted and statistically insignificant throughout: the broad dollar index moves $+0.06\%$ on day one and $+0.94\%$ over five days, neither distinguishable from zero (Table 5.1, Panel B). Crude rose meaningfully (Brent $+11.21\%$ at five days), but the currency response is essentially flat. This is the “ambiguous actor” case anticipated in the introduction: with the United States a co-belligerent rather than a pure safe harbour, the flight-to-the-dollar reflex and the erosion pressure roughly offset, leaving no significant net move.

5 Event Study Results

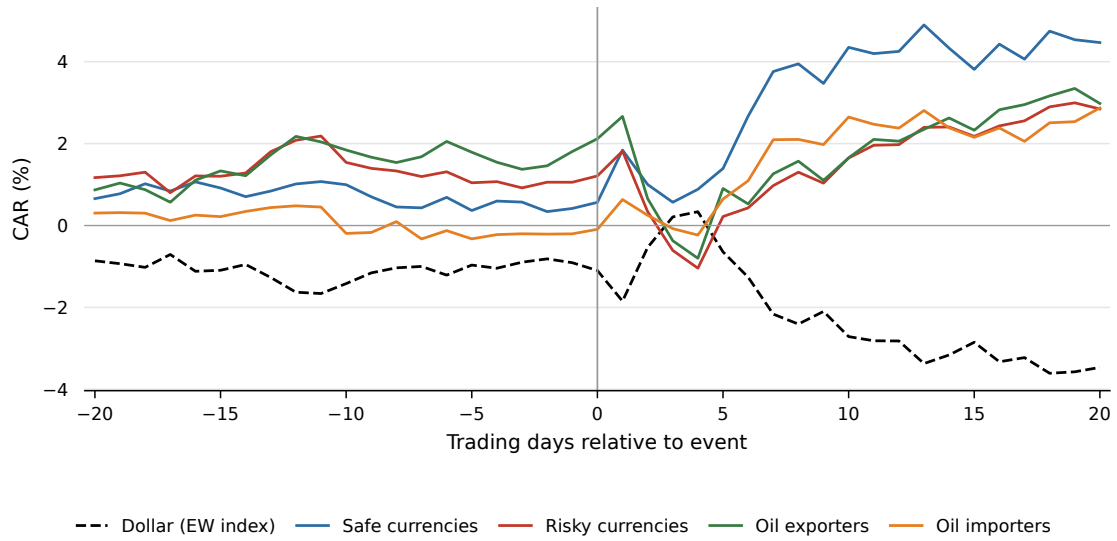


Figure 5.1: Liberation Day (2 April 2025): cumulative abnormal returns of the dollar and the currency baskets over ± 20 trading days. The dollar slides while the named-haven currencies bid; crude and European equities fall.

Hormuz crisis (February–March 2026). Here the dollar behaves as a classical safe haven. The broad index appreciates 1.40% on impact, building to +2.12% at ten days and +3.29% at twenty (Table 5.1). Crude spikes violently—Brent +27.83% and WTI +34.71% over five days—and the currency cross-section sorts on oil exposure: the oil-importer basket (yen, won, rupee) depreciates 1.10% against the surging dollar, underperforming the oil-exporter basket, which falls only 0.71% as higher crude cushions exporters. The dollar is bid even though the United States is the originator of the strikes, indicating that for a security/oil-supply shock the safe-haven reflex survives U.S. implication—in sharp contrast to the trade-policy shock.

Crucially, the Hormuz appreciation is *transitory*. By fifty days the broad-index CAR has returned to -0.13% (insignificant): the entire move unwinds. The clearest evidence that this was a genuine but temporary risk premium comes from the **April 8, 2026 ceasefire**, a within-crisis sub-event marked at +27 days in Figure 5.3. On the two-week ceasefire announcement—which committed to reopening the Strait of Hormuz—the dollar gave back 1.18% on the day and 1.63% over five days, and crude collapsed roughly 24% within ten days. The premium that the threat put *into* the dollar came back *out* on de-escalation, mapping the exchange-rate move directly onto the geopolitical risk it priced. This round-trip is about as close to clean identification as a single-country event study allows.

5.3 Cross-Event Comparison

Reading across Table 5.2, the two U.S.-originated shocks produce a *mirror image*. The trade-policy shock weakens the dollar and depresses oil; the military/oil-supply shock strengthens the dollar and lifts oil. The dollar is therefore not a fixed safe haven or a fixed casualty—its response is conditioned by the *type* of shock, even holding the originator (the United States) fixed.

This contrast is not merely descriptive. Table 5.3 tests the difference in the dollar’s CAR between events directly. At the ten-day horizon the broad dollar index is 4.93 percentage points lower under the tariff shock than under the Hormuz crisis ($t = -3.7$) and 4.51 points lower than under Ukraine ($t = -3.4$), both significant at the 1% level; by twenty days the tariff–Hormuz gap widens to 6.9 points. Crucially, the Hormuz and Ukraine responses are statistically *indistinguishable* (differences of 0.4–2.4 points, none significant): the dollar appreciates after the U.S.-originated Hormuz shock by as much as it does after the external Ukraine shock. The mirror image is thus driven entirely by the trade-policy shock—the only episode that moves the dollar significantly differently from a classical safe-haven event. The divergence is a horizon phenomenon: at five days no pair is yet significant, consistent with the announcement reaction taking up to two weeks to resolve.

Table 5.3: Difference in dollar CARs across events (broad USD index, %)

Contrast (A – B)	$\Delta\text{CAR}(0, h)$, percentage points		
	$h=5$	$h=10$	$h=20$
Liberation Day – Hormuz crisis	–1.54	–4.93***	–6.92***
Liberation Day – Ukraine	–1.06	–4.51***	–4.56**
Hormuz crisis – Ukraine	+0.48	+0.42	+2.35
Liberation Day – Twelve-day war	–0.94	–2.78*	–4.17*

Notes. Difference between two events’ cumulative abnormal returns of the broad nominal U.S. dollar index (%), constant-mean model, estimation window $[-130, -11]$. Standard error $\sqrt{L(\sigma_A^2 + \sigma_B^2)}$ with $L = h + 1$; events treated as independent (windows do not overlap). A positive value means the dollar rose more (or fell less) under shock A than shock B. */**/***: 10/5/1%.

The Ukraine benchmark disciplines the interpretation. As an external military/oil shock, it generates the textbook response—the dollar appreciates (+1.70% at ten days, +3.12% at fifty) and crude rises 13.63% (Table 5.2)—and, as just shown, that response is statistically the same as Hormuz. Two features nonetheless distinguish the external from the U.S.-originated military shock. First, the Ukraine appreciation

is durable whereas the Hormuz appreciation reverts, consistent with an external shock conferring a lasting safe-haven bid and a U.S.-implicated one only a temporary one. Second, and most importantly for the thesis, the *trade-policy* shock breaks the pattern entirely: the only episode in which the dollar depreciates is the one the United States itself authored on the economic-policy front. The safe-haven function is intact for geopolitical/oil-supply shocks—even U.S.-originated ones—but it inverts for a U.S. trade-policy shock.

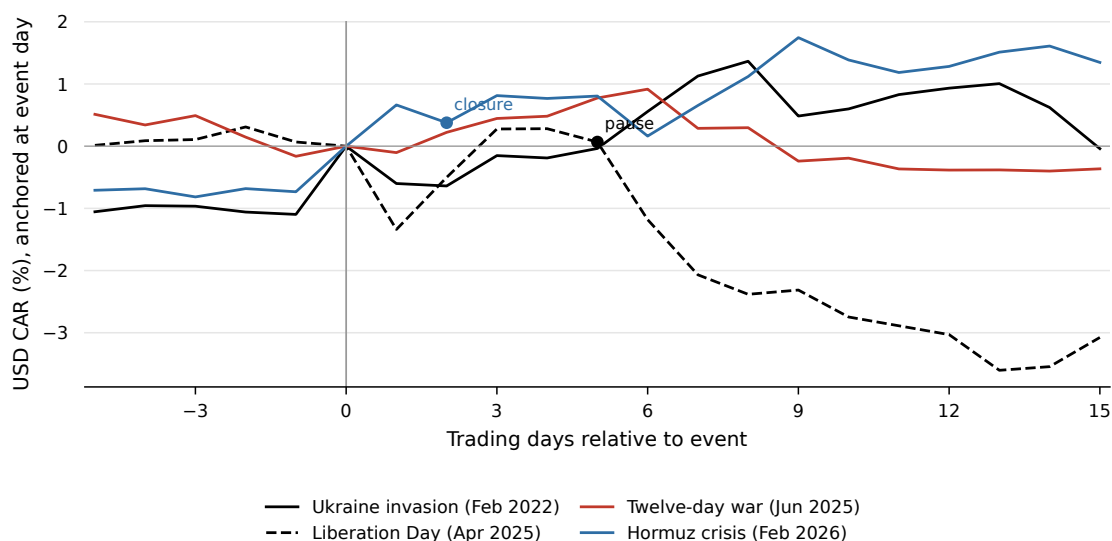


Figure 5.2: Broad dollar CARs across the four headline events, re-anchored on the event day (−5 to +15). The tariff shock is the only episode in which the dollar depreciates; Hormuz and Ukraine trace the classical appreciation.

5.4 Gold and Commodities

The commodity dimension reinforces the demand-versus-supply reading. Brent and WTI move in opposite directions across the two shock types—down $\sim 14\%$ on the tariff shock (a demand shock) and up $\sim 28\text{--}35\%$ on the Hormuz supply shock—and the two benchmarks agree closely, so the result does not depend on the choice of crude reference (Section 5.5 returns to this). The oil response is the channel through which the cross-section of oil-exporter and oil-importer currencies sorts in Table 5.2.

Gold tells a subtler story than the “universal haven” a first intuition would suggest. Rather than rising around every shock, gold tracks the *inverse of the dollar* across the U.S.-originated episodes: it gains when the tariff shock weakens the dollar ($+5.99\%$ at ten days, and $+8.48\%$ around the tariff pause) and falls when the Hormuz shock

strengthens it (-8.36% at ten days, deepening at longer horizons). The one departure is the external control: around the Ukraine invasion gold and the dollar rise *together* ($+4.24\%$ for gold at ten days), the dual-haven response a classical risk-off episode produces. Gold is thus not an unconditional substitute for the dollar but a hedge against it—for the U.S.-made shocks it moves opposite the dollar, reinforcing that the dollar’s own conditional response, and not a generic flight to safety, is the organising variable of this chapter; only for the external shock do both havens bid at once. Two caveats temper the magnitudes: gold appreciated roughly 240% over the sample, so the constant-mean baseline is unusually sensitive and the long-horizon gold CARs (e.g. Hormuz at twenty days) overstate the move—the *sign* is the robust feature, not the level—and the single-contributor quote used for the gold fix (LSEG XAU=ZKBZ) is noted in Appendix 10.

Equities corroborate the risk-off reading without the dollar’s conditionality. The Euro Stoxx 50 falls around every shock— -14.58% on Liberation Day, -8.23% around Hormuz, and -6.03% after the Ukraine invasion (all significant at 5% or better)—so European risk assets sell off whether the shock is a trade war, an oil-supply scare, or an external invasion. The contrast with the dollar is the point: equities are an *unconditional* casualty of geopolitical and policy shocks, whereas the dollar’s sign flips with the type of shock. The safe-haven question is therefore specific to the dollar, not a generic property of risk assets.

5.5 Robustness

Three checks support the readings above. First, *longer windows* confirm the persistence pattern (Table 5.1, $h=50$): the tariff depreciation deepens to -6.34% , the Hormuz appreciation fully reverts, and the Ukraine appreciation persists at $+3.12\%$ —the long horizon thus separates a level re-rating of the dollar (tariffs) from a transitory risk premium (Hormuz). The ± 50 -day figures (e.g. Figure 5.3) plot crude on a secondary axis and mark neighbouring sub-events, making both the persistence and the ceasefire reversal visible. Second, *Brent versus WTI*: the crude responses are quantitatively similar across all events (Table 5.2), so conclusions are invariant to the benchmark. Third, the *market model* with the dollar factor, reported for individual currencies in Appendix 10, leaves the safe-haven hierarchy unchanged.

Two limitations bound the long-horizon results. The ± 50 -day windows of the two 2025 events overlap (Liberation Day’s tail reaches the twelve-day war ~ 52 trading days later), so their long-horizon CARs are not fully independent; this is why the

5 Event Study Results

persistence figures are read event by event, while the formal cross-event *difference* tests (Table 5.3) use horizons up to twenty days, at which the windows of the compared pairs do not overlap. And the two most recent shocks—the May 2026 U.S. strikes and the June 2026 ceasefire—are omitted from the longer horizons because the sample ends 2026-06-16, leaving too few post-event trading days for a stable window.

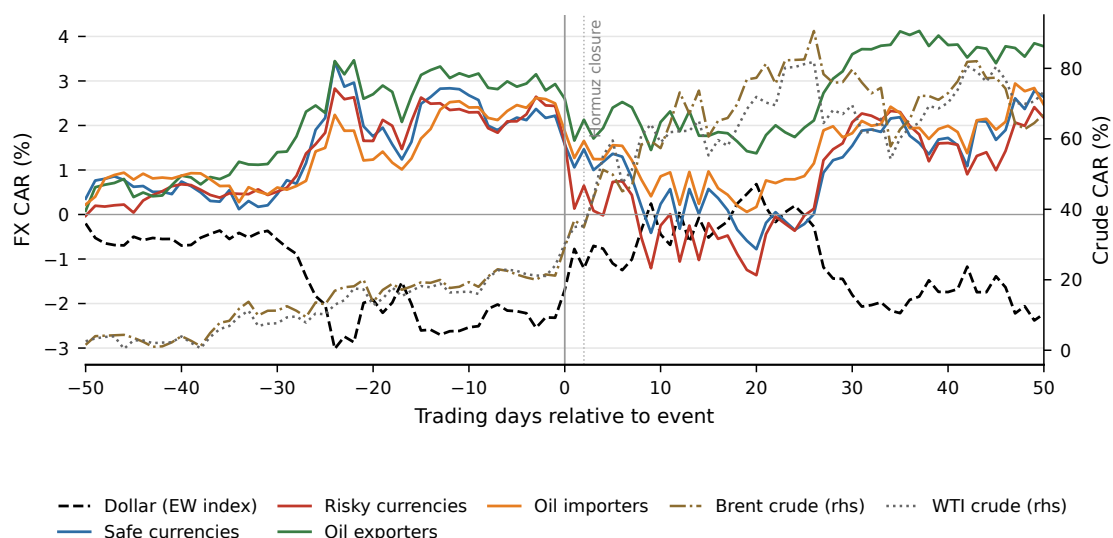


Figure 5.3: Hormuz crisis, ± 50 trading days. Left axis: currency-basket CARs; right axis: Brent and WTI CARs. The dollar’s appreciation reverts by +50; the April 8 two-week ceasefire (marked) both crashes crude and unwinds the dollar premium.

6 A Regression Test of the Safe-Haven Breakdown

The event study of Chapter 5 shows *that* the dollar moved in opposite directions across the two shock types. This chapter asks the sharper question directly: did the dollar’s defining safe-haven property—appreciating when global risk rises—actually hold or break in each episode? The dummy-interaction regression specified in Section 4.3 answers it with ordinary least squares and heteroskedasticity-robust standard errors, requiring none of the modelling assumptions of a conditional-correlation (GARCH) approach.

Table 6.1: Did the dollar’s safe-haven reflex survive each shock? (OLS, HAC standard errors)

	Coef.	t	Within-window slope $b_1 + \beta_k$
ΔVIX (normal, b_1)	+0.0003***	7.0	—
× Tariff (β_{tariff})	−0.0003***	−2.6	≈ 0.0000
× Hormuz (β_{Hormuz})	+0.0003	1.3	+0.0006
× Ukraine (β_{Ukraine})	+0.0008***	4.8	+0.0011
<i>Window (level) dummies</i>			
Tariff window	−0.0015*	−1.7	
Hormuz window	+0.0011**	2.1	
Ukraine window	+0.0010**	2.1	

Notes. OLS of the daily broad-dollar return on the daily change in the VIX, its interactions with crisis-window dummies, and the window dummies; Newey–West (HAC, 5 lags) standard errors. Crisis windows run from the event day to +20 trading days. $N = 1,944$, $R^2 = 0.06$. A positive coefficient on ΔVIX means the dollar appreciates when risk rises (safe-haven behaviour); the near-zero within-window slope for the tariff episode is the breakdown. */**/***: 10/5/1%.

The result is unambiguous (Table 6.1). On normal days the dollar is a safe haven: the coefficient on the change in the VIX is positive and strongly significant ($b_1 = 0.0003$, $t = 7.0$), so a risk-off jump in volatility lifts the dollar. During the **tariff** window this reflex *disappears*: the interaction is equal and opposite ($\beta_{\text{tariff}} = -0.0003$, significant at the 1% level), so the within-window sensitivity $b_1 + \beta_{\text{tariff}}$ is essentially zero. The dollar stopped behaving as a haven precisely when the United States was the source of the shock. During the **Hormuz** window the interaction is small and statistically insignificant ($t = 1.3$), leaving the safe-haven relationship unchanged from normal; during the **Ukraine** window it is, if anything, *stronger* ($\beta = +0.0008$,

$t = 4.8$). The window (level) dummies echo the event study—the average dollar drift is negative in the tariff window and positive in the Hormuz and Ukraine windows.

This regression reaches the event study’s conclusion by a different and fully transparent route: the dollar’s safe-haven function broke down for the U.S. trade-policy shock but remained intact for the geopolitical and oil-supply shocks. Because it rests on a single, interpretable least-squares specification with robust standard errors, it is the test most directly tied to the central hypothesis—and it delivers that test at a fraction of the modelling and estimation risk of a time-varying-volatility alternative.

7 Discussion

This chapter draws the event-study and regression results together, relates them to the literature, and sets out their implications and limits.

7.1 The Dollar’s Safe Haven Is Conditional on Shock Type

The central finding is that the dollar’s safe-haven response is conditioned by the *type* of shock, not merely by who causes it. The dollar appreciated around both military/oil-supply shocks—the external Ukraine invasion and the U.S.-originated Hormuz crisis—and the formal difference test cannot distinguish the two responses, even though the United States authored one and merely witnessed the other. Yet it depreciated around the one purely economic-policy shock it created, the Liberation Day tariffs. Origin alone does not explain the pattern; the nature of the shock does. This refines the “dollar-erosion” reading ([Ostry, Lloyd and Corsetti 2025](#); [Kamin 2025](#)): the safe-haven function is not weakening in general but is specifically suspended when the United States is itself the source of an economic-policy shock, while it survives intact—reversion notwithstanding—for the security shocks that the safe-haven tradition has always emphasised ([Ranaldo and Söderlind 2010](#); [Habib and Stracca 2012](#)).

The regression sharpens this further. Estimated over the full sample, the dollar’s sensitivity to risk is significantly positive on normal days—the textbook safe-haven reflex—but the interaction term shows it collapsing to essentially zero inside the tariff window, while it remains intact during the Hormuz and Ukraine windows. The breakdown is thus not merely a difference in the *level* of the dollar across episodes but a change in its very *relationship with risk*: during the U.S. trade-policy shock the dollar stopped trading like a safe haven, and only then.

7.2 Gold and the Oil Channel

Gold does not behave as the universal safe haven that a first reading of [Baur and Lucey 2010](#) would predict. Instead it tracks the inverse of the dollar across the U.S.-originated shocks—rising when the tariff shock weakened the dollar, falling when the Hormuz shock strengthened it—and bids alongside the dollar only for the

external Ukraine shock. Gold is thus better described as a hedge *against* the dollar than as a substitute for it, which reinforces, rather than competes with, the chapter’s main message: the dollar’s own conditional move is the organising variable, and gold prices it from the other side.

The oil channel operates as the commodity-currency literature predicts, but only for the supply shock. During the Hormuz crisis, oil-importer currencies underperformed oil-exporters as crude surged, consistent with [Chen and Rogoff 2003](#) and [Kilian 2009](#); during the demand-driven tariff decline the sorting was muted. The yen illustrates the resulting tension: a classical safe haven that is also a large oil importer, it is pulled in opposite directions during an oil-supply shock, which is part of why the basket responses around Hormuz are smaller and noisier than the dollar-index response.

7.3 Policy Implications

Three implications follow. For reserve managers and central banks, the dollar’s safe-haven status cannot be treated as unconditional: hedging programmes calibrated on the assumption that the dollar always rallies in risk-off episodes will misfire precisely when the shock originates in U.S. economic policy. For portfolio managers, diversification should be thought of across *shock types*, not only across assets—gold and competing-haven currencies, not the dollar, carried the flight to quality in the tariff episode. For the broader debate on the dollar’s reserve role ([Obstfeld and Zhou 2023](#); [Jiang et al. 2026](#)), the evidence is a caution against over-reading any single episode: the 2025 tariff shock did dent the dollar, but the 2026 security shock showed the haven bid still firmly in place.

7.4 Limitations

Several caveats bound the results. The two most recent events—the May 2026 U.S. strikes and the June 2026 ceasefire—have too few post-event trading days for the longer windows, so they are reported only at short horizons. The regression’s crisis windows are defined around the events rather than by a structural break test, and the VIX–dollar link it measures is a reduced-form association, not a causal channel. Gold’s exceptional 2025–26 trend makes its constant-mean CARs sensitive in level, so they are read for sign rather than magnitude. As a single-country event study around a small number of recent, partly overlapping episodes, the design trades

7 Discussion

breadth for the clean U.S.-versus-external contrast; and without forward/basis data beyond the one-month tenor, covered-interest-parity deviations are left for future work.

8 Conclusion

This thesis asked whether the U.S. dollar’s safe-haven response depends on the type of shock that triggers a crisis, even when the United States is implicated in the shock itself. Two recent episodes provided an unusually clean test: the 2025 Liberation Day tariffs, a purely U.S. economic-policy shock, and the 2026 Iran/Hormuz crisis, a geopolitical oil-supply shock, with the 2022 Russian invasion of Ukraine as an external control.

The evidence gives a clear answer. The two shock types produced a mirror image. The dollar depreciated around the tariff shock—roughly four percent on the broad index within twenty trading days—while it appreciated around the Hormuz crisis and around Ukraine, and a formal difference test confirms the tariff response is significantly distinct from both security episodes, which are themselves statistically indistinguishable from one another. The dollar’s haven status is therefore conditional on the *nature* of the shock, not on its origin: it survives a U.S.-originated security shock but inverts for a U.S.-originated economic-policy shock. A complementary regression confirms the breakdown statistically: the dollar’s safe-haven sensitivity to risk—significantly positive on normal days—collapses to zero in the tariff window, while holding through the Hormuz and Ukraine episodes. Two further results round out the picture: gold acts as a hedge against the dollar rather than as a universal safe haven, and the oil channel sorts currencies only for the supply-driven shock.

The thesis contributes the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets, carries a gold and oil benchmark through the same design to connect the safe-haven and commodity-currency literatures, and extends standard cross-sectional currency tools to mid-2026 on a consistent WMR dataset.

Several extensions suggest themselves. As the geopolitical situation continues to evolve, the sample can be extended so that the most recent episodes acquire full post-event windows. Richer forward and basis data would allow covered-interest-parity deviations to be brought into the analysis, and intraday data would tighten the identification around the announcement windows. A dynamic conditional-correlation (DCC-GARCH) model of the dollar–VIX relationship would test the safe-haven-breakdown hypothesis directly and is the natural next layer. Finally, a formal model of shock-dependent safe-haven demand—in which the currency of the shock’s originator loses its haven premium for policy shocks but not for security shocks—would give the empirical pattern documented here a theoretical foundation.

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10 Data Sources

All data are daily, 1 January 2019 to 30 June 2026. The macro-financial block (dollar indices, Treasury and TIPS yields, breakeven inflation, the VIX, Brent and WTI crude, Henry Hub gas, and the S&P 500) is from FRED. The 32-currency spot and one-month forward panel, gold, the Euro Stoxx 50, and the TTF gas contract are exported from LSEG at the WM/Reuters fixing. The Geopolitical Risk index is from Caldara and Iacoviello, and the Economic and Trade Policy Uncertainty indices from *policyuncertainty.com*. The analysis pipeline (`consolidate_lseg.py` → `get_data.py` → `build_fx_factors.py` → `02_build_returns.py` → the event-study scripts 03–05 and the safe-haven regression `07_safehaven_regression.py`) reproduces every table and figure in the thesis from these sources.

Appendix

1 Currency Universe and Classification

Table 1 lists the 32 currencies in the panel with their carry-sorted bucket (low/middle/high forward-discount tercile, pegged and managed currencies excluded from the sort) and their oil-trade role. The named safe-haven robustness baskets used in Chapter 5 are the yen, Swiss franc and euro (safe) against the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah (risky).

2 Robustness

Market model. Alongside the constant-mean specification, the event study computes market-model abnormal returns for the individual currencies, using the equal-weighted dollar factor as the market return (`Code/03_event_study.py`, method `market_model`). These are reported in the replication output; the safe-haven ordering across currencies is unchanged relative to the constant-mean results in Chapter 5.

Regression robustness. The safe-haven regression of Chapter 6 uses Newey–West (HAC) standard errors, so its inference is robust to the autocorrelation and volatility clustering of daily returns—the main reason a simple least-squares interaction is preferred here to a conditional-correlation model. The crisis windows are the same event-day-to-+20 windows used in the event study, so the regression and the event study rest on a common, transparent definition of each episode.

Table 1: Currency classification

Code	Currency	Carry bucket	Oil role
AUD	Australian dollar	middle	—
BRL	Brazilian real	high (risky)	exporter
CAD	Canadian dollar	middle	exporter
CHF	Swiss franc	low (safe)	—
CLP	Chilean peso	no 1M forward	—
CNY	Chinese yuan	pegged/managed	—
COP	Colombian peso	no 1M forward	exporter
CZK	Czech koruna	middle	—
DKK	Danish krone	pegged (EUR)	—
EUR	Euro	low (safe)	—
GBP	Pound sterling	middle	—
HKD	Hong Kong dollar	pegged (USD)	—
HUF	Hungarian forint	high (risky)	—
IDR	Indonesian rupiah	high (risky)	—
ILS	Israeli shekel	low (safe)	—
INR	Indian rupee	high (risky)	importer
JPY	Japanese yen	low (safe)	importer
KRW	South Korean won	middle	importer
KWD	Kuwaiti dinar	pegged	exporter
MXN	Mexican peso	high (risky)	exporter
MYR	Malaysian ringgit	middle	—
NOK	Norwegian krone	middle	exporter
NZD	New Zealand dollar	middle	—
PEN	Peruvian sol	no 1M forward	—
PLN	Polish zloty	high (risky)	—
SAR	Saudi riyal	pegged	exporter
SEK	Swedish krona	low (safe)	—
SGD	Singapore dollar	low (safe)	—
THB	Thai baht	low (safe)	importer
TRY	Turkish lira	high (risky)	importer
TWD	Taiwan dollar	low (safe)	importer
ZAR	South African rand	high (risky)	—

Notes. Carry buckets are terciles of the average one-month forward discount (Section 4.1.3); pegged/managed currencies and the three currencies without a clean one-month forward (CLP, COP, PEN) are excluded from the carry sort. Oil role follows the net oil-trade classification.

Affidavit

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